

Artificial Intelligence and Machine Learning: An Introduction

1. Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks commonly associated with human intelligence. These tasks include reasoning, learning from experience, understanding language, recognizing patterns, making decisions, and solving problems. AI systems are used in areas such as healthcare, finance, education, transportation, cybersecurity, and customer service.

2. Machine Learning

Machine Learning (ML) is a major branch of artificial intelligence. Instead of programming every decision explicitly, machine learning algorithms learn patterns from data and use those patterns to make predictions or decisions. Common learning approaches include supervised learning, unsupervised learning, and reinforcement learning.

3. Supervised and Unsupervised Learning

In supervised learning, a model learns from labeled examples. Classification predicts a category, while regression predicts a numerical value. In unsupervised learning, the data does not contain predefined target labels. Algorithms may discover groups, relationships, or lower-dimensional representations in the data. Clustering is a common unsupervised learning technique.

4. Deep Learning

Deep Learning is a subset of machine learning based on artificial neural networks with multiple layers. Deep learning has been especially successful in computer vision, speech recognition, natural language processing, and generative AI. Convolutional neural networks are commonly associated with image processing, while transformer architectures are widely used for modern language models.

5. Natural Language Processing

Natural Language Processing (NLP) enables computers to process and generate human language. Applications include sentiment analysis, translation, text summarization, information extraction, search systems, and conversational assistants. Modern NLP systems frequently use transformer-based models to represent and generate language.

6. Large Language Models

Large Language Models (LLMs) are neural language models trained on large collections of text. They can generate text, answer questions, summarize information, assist with coding, and perform many other language tasks. However, an LLM may sometimes generate information that is inaccurate or unsupported, so applications often use additional techniques to provide reliable context.

7. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with a generative language model. A typical RAG system first divides documents into smaller text chunks. Each chunk is converted into a numerical embedding and stored in a vector database. When a user asks a question, the question is also converted into an embedding. The system retrieves the most semantically similar chunks and provides them as context to the language model, which then generates an answer grounded in the retrieved information.

8. Embeddings and Vector Search

An embedding is a numerical vector that represents the meaning of text or other data. Semantically similar pieces of text tend to have embeddings that are close to each other in vector space. Vector search systems such as FAISS can efficiently compare embeddings and retrieve relevant document chunks for a user's query.

9. Example RAG Workflow

Consider a chatbot that answers questions about a PDF. First, the application extracts text from the PDF. Second, it splits the text into overlapping chunks. Third, it creates embeddings for those chunks and stores them in a vector index. Fourth, a user question is embedded and compared with the stored vectors. Finally, the most relevant chunks are sent with the question to a language model, which produces a context-aware answer.

10. Benefits and Limitations

RAG can make AI assistants more useful for private or domain-specific documents because the model receives relevant information at query time. It can also reduce unsupported answers when the prompt requires the model to rely on retrieved context. The quality of a RAG system still depends on document quality, chunking strategy, embedding quality, retrieval accuracy, and the instructions given to the language model.

Conclusion

Artificial intelligence includes a broad range of techniques for building intelligent computer systems. Machine learning enables systems to learn from data, while deep learning and large language models support increasingly capable applications. Retrieval-Augmented Generation connects language models with external documents through embeddings and vector search, making it a

practical architecture for document question-answering chatbots.